Research into how the human body digests and further metabolizes ingested foods is critical to understanding human health processes and diseases, such as obesity and diabetes. However, researchers interested in studying the human digestive system must overcome many obstacles, including ethical constraints in using human and animal subjects, the high cost of experiments, and the lack of standardized conditions across human subjects. Because of these obstacles, researchers began designing and using *in vitro* digestive system models in the 1990s (Figure 1).

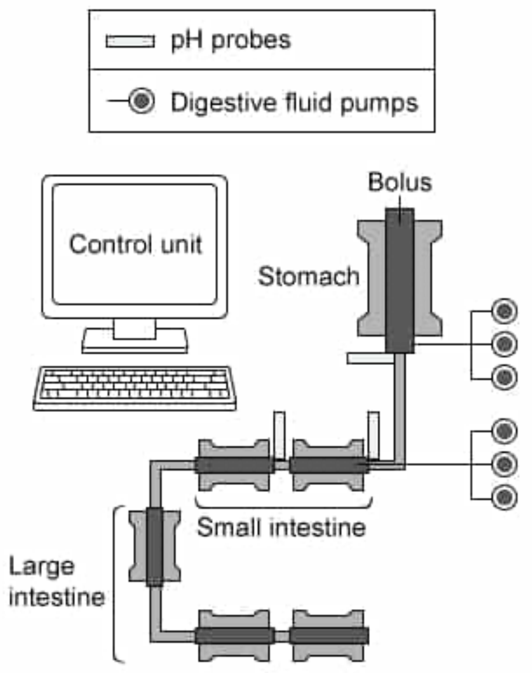


Figure 1 Example of *in vitro* digestive system model

There are two basic *in vitro* models of the human digestive system: static models and dynamic models, both of which have advantages and disadvantages. Static models, which make use of a series of glass chambers that model digestive organs, are more cost effective and practical for large sample sizes; however, they do not mimic the mechanical force or dynamic conditions produced in humans. While dynamic models are more costly, these model types come closer to approximating *in vivo* conditions, including mechanical, dynamic, and/or chemical conditions.

Regardless of model type, researchers employing *in vitro* digestive system models use a complex system of computer controls to adjust for variables such as pH, transit time of food particles, and digestive secretions.

Many *in vitro* digestive system models lack representations of accessory digestive organs such as the liver. Which of the following products must be added to these systems by researchers to simulate the liver's contribution(s) to *in vivo* conditions in the duodenal lumen?

I. Insulin
II. Glucose

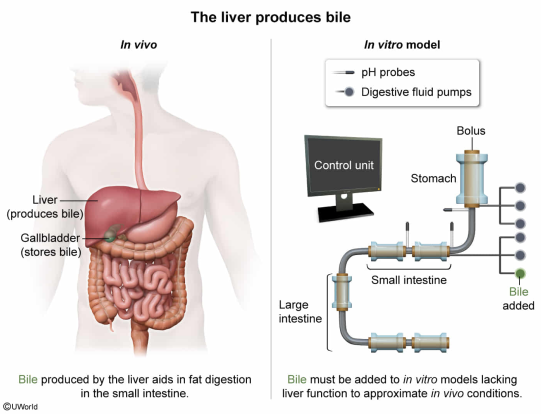
III. Bile

◯ A. I only [3%]

- [x] B. III only [58%]
- C. I and II only [10%]
- D. II and III only [27%]

Correct
58% answered correctly

Explanation:



The liver is an organ responsible for several functions, including blood glucose regulation, macromolecule synthesis (eg, plasma proteins), storage of molecules (eg, glycogen, iron), production of bile, and detoxification of drugs and waste products (eg, bilirubin).

This question states that many *in vitro* digestive system models lack accessory digestive organs such as the liver. Because of this, any products of the liver used in digestion must be added to *in vitro* models to simulate *in vivo* conditions.

One of the products of the liver is bile, a solution that, once it has been produced, is stored in the gallbladder and released into the duodenum of the small intestine to help mechanically (ie, by nonenzymatic and nonchemical means) digest lipids. Bile salts, one of the components of bile, break down large lipid globules into micelles (smaller droplets) during emulsification. Therefore, **bile** is a product used in the duodenal lumen that must be added by researchers to simulate *in vivo* conditions in the duodenal lumen **(Number III)**.

(Number I) Insulin is a product of pancreatic beta cells (*not* the liver). Insulin promotes glucose uptake by certain tissues (eg, muscle, adipose), decreases liver glucose production, and increases glucose storage in glycogen molecules (ie, glycogenesis).

(Number II) While the liver can produce glucose via gluconeogenesis or glycogenolysis, liver glucose is *not* released into the duodenal lumen during digestion.

Educational objective:

Bile is a substance synthesized by the liver and stored by the gallbladder. When released into the duodenal lumen, it aids in the mechanical digestion of lipids. Other functions of the liver include blood glucose regulation, macromolecule synthesis, molecular storage, and detoxification.

Time spent:0 sec	**QID:**403883
Subject: Biology	**System:** Digestion and Excretion

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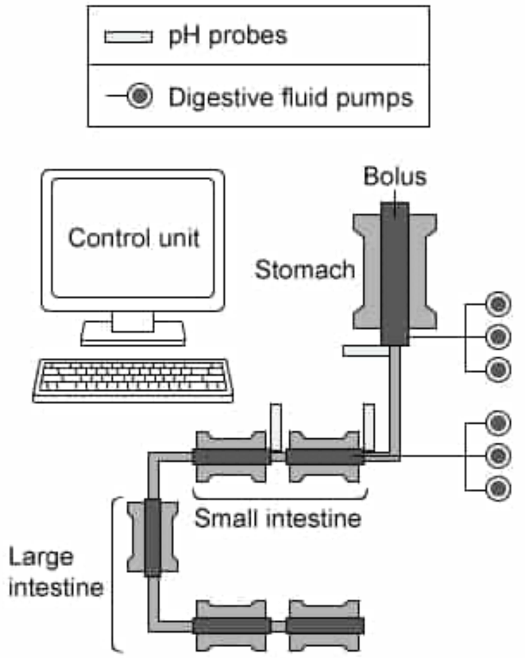

Figure 1 Example of *in vitro* digestive system model

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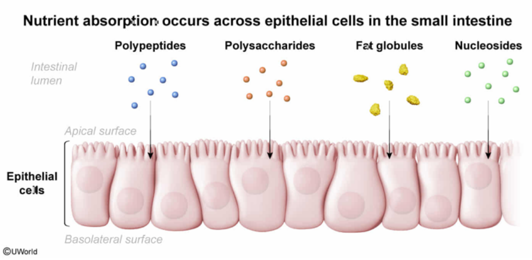
Regardless of model type, researchers employing *in vitro* digestive system models use a complex system of computer controls to adjust for variables such as pH, transit time of food particles, and digestive secretions.

A filter mechanism is used in some *in vitro* digestive system models. Why are these filter mechanisms helpful to investigate digestive conditions *in vivo*?

- A. To model movement of nutrients across small intestine epithelial cells [78%]
- B. To model lipid absorption into blood vessels in the large intestine wall [7%]
- C. To model filtration of water across epithelial cells lining the esophagus [4%]
- D. To model filtration of proteins across the stomach lining [9%]

Correct
78% answered correctly

Explanation:



The action of peristalsis moves ingested food from the mouth through the gastrointestinal tract, which consists of several components.

1. The esophagus is a hollow tube that moves ingested food from the mouth to the stomach as a bolus (ie, compact food mass).

2. In the stomach, the bolus is chemically and mechanically digested, creating a semifluid substance called chyme, which moves into the small intestine.

3. The small intestine is a long tube with three subdivisions, the duodenum, jejunum, and ileum, in which macromolecular digestion is completed and substances useful to the body are absorbed. Absorption occurs across the epithelial cells lining the small intestine. Any undigestible material passes to the large intestine.

4. In the large intestine, water and electrolytes are absorbed from undigestible material to form solid feces.

This question states that a filter mechanism is used in some *in vitro* digestive system models and asks why these mechanisms are helpful to investigate digestive system conditions *in vivo*. In the **small intestine, nutrients move across epithelial cells** during absorption by the body. Therefore, to successfully model *in vivo* conditions, *in vitro* models may use filter mechanisms to remove materials (useful nutrients) from the material in the model lumen.

(Choice B) Absorption of lipids and other macromolecules is complete by the time ingested food reaches the end of the small intestine. Subsequently, in the large intestine, water and electrolytes (*not* lipids) are absorbed. Therefore, a filter mechanism should not be used to model large intestine lipid absorption *in vitro*.

(Choice C) The esophagus serves only to move ingested food from the mouth to the stomach. No absorption or digestion occurs in the esophagus. Consequently, a filter mechanism within an *in vitro* digestive system model should not be used to model filtration across epithelial cells lining the esophagus.

(Choice D) While proteins undergo chemical and mechanical digestion in the stomach, proteins

are *not* filtered across the stomach lining. Therefore, a filter mechanism should not be used to model filtration of proteins across the stomach lining.

Educational objective:
The small intestine, consisting of the duodenum, jejunum, and ileum, is responsible for the digestion of food and absorption of useful nutrients across epithelial cells.